

Towards a Reference Architecture for Demand-Oriented Public Transportation Services

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Abstract—New mobility services (NMS) with innovative business models become more and more popular in cities and urban areas. This paper intends to make a scientific contribution to the field of enterprise architecture management in public transport companies. Previous work discovered the need for a recommendation or a standardized approach for the NMS integration into EA. Reference architectures intend to offer such recommendations, but the analysis of existing work showed that established reference architectures do not cover NMS sufficiently. Thus, the focus of this work is (a) examination of the feasibility of extending public transportation reference architectures and (b) developing the initial design of such an extension based on a complex real-world case. A demand-oriented in addition to supply-oriented approach to public transportation is one of the proposed features of the reference.

Keywords—New Mobility Services, Digital Transformation, Enterprise Architecture, Reference Architecture, Case Study, Demand-oriented transportation services

I. INTRODUCTION

New mobility services with innovative business models become more and more popular in cities and urban areas. City bikes, e-scooters or bike sharing that are offered with pay-as-you-go or subscription models are well-known examples and expected to affect the way how people move within short and medium distances in larger cities [1]. These new or innovative mobility solutions (NMS) and have been subject to research, for example regarding their business models [2] and factors affecting service offerings [3]. However, their influence on conventional public transportation (PT) and the IT-landscape or enterprise architecture of PT companies is still subject to research.

This paper intends to make a scientific contribution to the field of enterprise architecture management in public transport companies. In previous work, the challenges caused by NMS for public transportation enterprises were investigated, the problems to be tackled in scientific research more precisely defined [4] and the impact on enterprise architectures [5] examined. One conclusion of our previous work was the need for a recommendation or a standardized approach for the NMS integration into EA. Reference architectures intend to offer such recommendations, but the analysis of existing work showed that established reference architectures do not cover NMS sufficiently. Thus, the focus of this work is (a) examination of the feasibility of extending public transportation reference architectures and (b) developing the

initial design of such an extension based on a complex real-world case.

This paper examines the current situation of NMS in the largest German public transport company, Berliner Verkehrsbetriebe AöR (BVG), using the example of the ride-pooling or on-demand ridesharing service BerlKönig as a case study (see section IV.). The reference architecture investigated and extended in this context is ITVU industry model (see section III.).

Another objective of this publication is to present an initial architectural model as a reference for the integration of NMS into a traditional public transport company. The starting point of this objective as well as of the present paper are the following statements, which have been presented in more detail in the previous work mentioned above:

1. There is a need for the integration of NMS into the current enterprise architecture of traditional public transport companies.
2. A reference architecture can support public transport companies in the development and integration of NMS in their own company.

As mentioned above, one aim of this paper is to first embed a specific part of BerlKönig into the enterprise architecture of a pipeline-oriented public transport company. Based on the information presented, the authors will answer the following central question:

- What could an integration of the BerlKönig service into a pipeline company look like based on a subarea selected as an example?

The paper is structured as follows: After a short introduction to the research method and theoretical background, the main subjects of this paper - mainly the case of the New Mobility Service BerlKönig, the ITVU industry model and the two types of public transport services from a digitalization perspective - a comparison between supply-oriented and demand-oriented public transport services is developed in the form of a delta analysis. Subsequently, the results of an interview about the integration of BerlKönig into the BVG are evaluated. Based on this, the authors show an example of a "simple" organizational-procedural and technical integration using the BerlKönig service and the functional area of sales. Finally, the results are summarized and evaluated and an outlook on a planned follow-up paper is given.

II. RESEARCH APPROACH

The work presented in this paper is part of a research aimed at method and tool support for integrating NMS into EA of public transportation organizations. It follows the paradigm of design science research [6]. This study concerns a step towards the initial design of the envisioned design artifact: EA method support for integrating NMS including tool support. The work presented in this paper started from the following research question: RQ: In the context of NMS introduction, what structural changes have to be implemented in typical EA of public transportation organizations?

The research method used for working on this research question is a combination of case study, conceptual modeling and argumentative-deductive work. Based on the research question, we started identifying "typical" EA for PT organizations by looking for reference architectures. Our intention was to find widely accepted and used architectures that can be considered as "industry standard" and examine them for change needs. Here, we identified the ITVU core model as one of the most frequently used in German PT organizations. Here, we draw two resources (cf. section III.): the ITVU industry model ("IT systems for transport companies") is a generic architecture developed by the company IVU for traditional public transport companies (see section III.A.), and the Open Mobility Platform (OMP) of the VDV Association.

For the integration of NMS into this ITVU and OMP architecture, we decided to use a real-world example and case study, which is the BerlKönig in the PT of Berlin. This included an interview with Michael Bartnik (Project Manager Jelbi, Berliner Verkehrsbetriebe) to analyze the details of integration of the Berlin ride-pooling service BerlKönig into BVG's corporate architecture. We modeled the integration and investigate to what extent the case study can serve as basis for generalization.

In a future step, we intend to develop an overall solution based on the requirements as they arise through a platform enterprise. Both pipeline and platform approaches (pipeline and platform organization [7]) will be investigated in a future version of the architecture.

III. BACKGROUND AND THEORETICAL FRAME

In preparation for the case study, the ITVU industry model is presented in section III.A. A general introduction EA and reference architectures forms the theoretical frame of the work (section IIIB.).

A. ITVU sector model (ITTC core model)

The ITVU industry model was developed and maintained by Gero Scholz, an employee of IVU Traffic Technologies AG, as a kind of simplified industry model of traditional public transport until 2016 [8]. The latest version of this model maybe found in the publication "IT systems in public transport", in which the domain model describes generic business processes on the one hand, and fundamental data structures (classes) of public transport on the other [9].

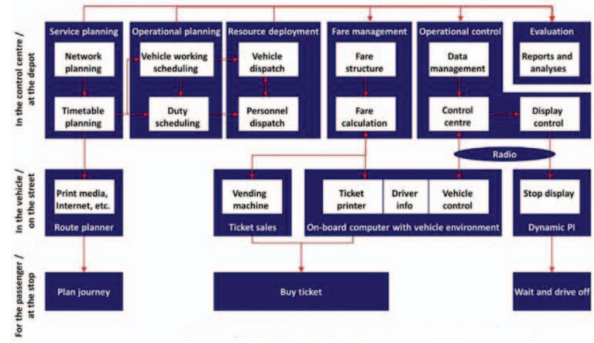


Fig. 1. Systems of the ITVU sector model [9]

In the ITVU model, 42 IT systems/applications are described in short profiles. These fact sheets contain:

- Names in the form: "plan.timetable",
- Designation ("Timetable"),
- brief summary of the function,
- Data flows to and from the individual applications in an overview,
- Users/actors in the form of roles.

Figure 2 shows the corresponding profile for the "Timetable planning" application.

Name	plan.timetable
Bezeichnung	Fahrplanung
Art	zentral
Nutzer	Fahrplaner
Funktionen	Beschreibung von Fahrwegen Definition von Linien Erfassen von Fahrzeitprofilen je Strecke und Tageszeit Festlegung von Fahrten, Zuordnung von Fahrzeugtypen Verwendung variabler Gültigkeitsmuster (Fahrplanperiode, Tagesart, einzelner Kalendertag) Zuordnung von Aspekten (Regelverkehr, Baustelle, Schülerverkehr, Werksverkehr) zu Fahrten Definition von Anschlüssen
Datenflüsse	→ plan.block Menge der Fahrten → pool.data Menge der Fahrten zur Abstimmung mit Verbundpartnern und zur Erstellung von Druckerzeugnissen → contract.management Fahrplandaten verbindliche Anschlüsse

Fig. 2. Application profile of the ITVU sector model [9]

Users and actors (roles) are classified in the ITVU sector model in 16 roles, which are described in detail with activity profiles. As an example, the description of the role "journey planner" is reproduced here in a profile text:

"The journey planner implements the traffic demand on the previously created network. Often the roles of network planner and timetable planner are not separated, as both activities are closely related. The timetable planner defines the journeys for each line (frequency and exact times) and takes connection relationships into account. He determines the type of vehicle to be used and takes restrictions that may result from the routing into account. In addition, any special equipment required, such as disabled access, is specified. Network and timetable planning can be the responsibility of

the transport company or can be specified in the transport contract, depending on with how much differentiation the public transport authority specifies the transport services to be provided [8].

Based on the applications shown in Figure 1 (ITVU industry model), the underlying data and process structures are assigned to the systems in a class model [9].

A general evaluation of business processes, such as accounting, human resources management or marketing, etc., is not comprehensively considered by the model - however, it offers a good introduction to special areas of the enterprise architecture of a public transport company.

B. EA and Reference EA

Enterprises active in public transportation need to be aware of the relations among their strategy, processes, applications and infrastructures as a basis to quickly react to changing demands in the market, such as the NMS integration. The Enterprise Architecture (EA) research domain contributes to this purpose by providing methods and tools to establish a more holistic perspective on enterprises [10][11]. This includes methods and procedural approaches to systematically capture an EA with modeling languages like the industry standard ArchiMate [12]. EA models represent different architectural layers of an organization, such as business, information and technology architecture. As EA projects are resource- and time-consuming, organizations usually have benefits from reference models for EA. A Reference Architecture is a generic EA for an application domain that is used in concrete enterprises as foundation in the design and realization of the concrete EA [13].

Many methods exist to develop reference models — both deductive and inductive approaches [14]. In general, reference models are developed for a certain problem in a certain application domain. The purpose is to be reused in a concrete application case in this domain. The reuse of a reference model is intended to increase both efficiency and effectivity of an enterprise's IT and change management [15]. For constructing reference models two general strategies are discussed in research: The deductive strategy derives reference models from generally accepted knowledge, the inductive approach abstracts from individual models to reach on a common understanding within the reference model [16].

IV. SUPPLY- AND DEMAND-ORIENTED OR PLATFORM- AND PIPELINE-ORIENTED PUBLIC TRANSPORT SERVICES

A. General Idea

In the course of the global digital transformation, demand-oriented public transport services are emerging alongside the traditional supply-oriented ones (see examples in 1. and 2.). For further consideration, the orientation of public transport services in the digital age is presented in the following:

Traditional public transport companies, such as the BVG in Berlin or the HVV in Hamburg, are primarily **supply-oriented** in their traditional orientation. This means that through their route planning and provision of the different modes of transport - bus, tram, underground and suburban railway - these companies make an **offer to** the population (to be transported) and the population uses this offer as required.

In contrast, most NMS (such as Uber) follow a **demand-driven approach**. This means that services such as ride-pooling (shared call taxi), car sharing or taxis are more

demand-driven and that this **demand** is actively made by the person (to be transported) by calling a shared taxi or looking for a free car sharing car.

The combination of these two different approaches gives rise to requirements which the authors will examine in more detail. Furthermore, the authors will shed light on the two paradigms which result from the business process orientation - namely a platform-oriented vs. a pipeline-oriented business model. At this point, it is safe to assume that platform-oriented business models are more likely to be found on the side of demand-driven public transport services and vice versa. This means that there is a correlation between **platform-oriented business models** that are **demand-oriented** (new public transport companies, digital disruption business models and processes, such as Uber) and **pipeline-oriented business models** that provide **supply-oriented public transport services** (traditional public transport companies, such as line operators).

The requirements resulting from these results are explained in the following section of this document. Examples will be given here and consolidated in a follow-up paper as well as modelled on the basis of the ITVU industry model. Hereby, the focus is on the distribution part of public transport companies for full integration.

B. Comparison between the functional requirements of demand- and supply-oriented public transport services

In this comparison, the authors will use an example to elaborate upon the differences between supply- and demand-oriented public transport services with regard to technical requirements. Even a preliminary review of the characteristics discussed herein turns up differences, which will be made concrete via an example below:

Operational planning and service planning

Differences between the two services in the areas of supply and operational planning are to be expected. The supply-oriented services plan tours on the basis of (stop-oriented) timetable planning. Vehicle scheduling on the one hand and work scheduling (part of operational planning) or respectively personnel scheduling on the other hand are based on this. There is a connection between the scheduled vehicle and the scheduled driver.

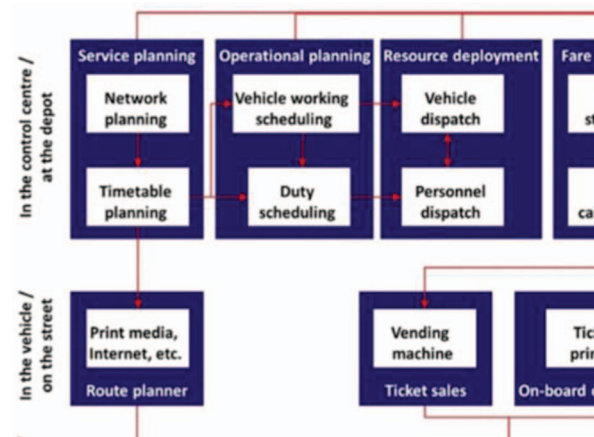


Fig. 3. Excerpt from the ITVU sector model for supply-oriented public transport services [9]

This complex process is not necessary for purely demand-responsive services, as the scheduled driver takes an appropriate vehicle from a vehicle pool or the sharing vehicle (car, e-scooter, etc.) is available for the user at a pick-up location. There is no need to keep extra tours ready for scheduled drivers and corresponding vehicles, as is the case with supply-oriented services. For this area, a separate standardized procedure must be developed, e.g. along the lines of the taxi companies.

Sharing vehicles drastically reduce the complexity in this area compared to supply-oriented public transport operations. However, this does not mean that sharing services do not require more complex tasks in other areas. For example, in the operational area (operational controlling), it must be ensured that the shared vehicles are continuously distributed optimally over the defined area of operation. This logistical task, which partly occurs at night, does not apply to services such as Uber, taxis, buses and trains, as these are moved and simultaneously "managed" by drivers.

This first rough comparison clarifies the functional requirements for an integrated public transport system, which takes into account both supply- and demand-oriented services. This analysis also illuminates the first signs of a higher degree of complexity required of the future EA (as an artefact).

V. NMS CASE STUDY BERLKÖNIG

A. Introduction to Berlin BerLKönig

The Berlin-based BerLKönig is a ride-pooling service which is being developed as a new mobility service, including digital infrastructure [19]. The service currently operates only in the variant 1 shown below:

BerLKönig A: Collective call taxi with free choice of destination in the eastern inner-city area of Berlin: fare zone A (inner Berlin S-Bahn ring).

BerLKönig A is operated in cooperation by the companies BVG and ViaVan, a subsidiary of the Daimler Group. ViaVan operates the service on behalf of BVG and supports management decisions by providing information.

In November 2017, the partners applied for approval for the BerLKönig concept. The service began operating in September 2018 [19].

BerLKönig BC: Collective call taxis with a fixed destination (nearest BVG underground station) and semi-variable route: Farezone BC (outer Berlin S-Bahn ring and the villages in the east of Berlin).

BerLKönig BC was launched by an initiative of the Berlin Taxi Guild in cooperation with BVG, ViaVan and Taxi Berlin. This federally funded service started operations in the summer of 2019. It was discontinued by BVG in May 2021 due to the cancellation of federal funding. Therefore, this variant is not considered in this paper [17].

B. Professional and IT integration of the BerLKönig

The current situation of the BerLKönig is determined via an expert interview with Michael Bartnik¹ of the BVG. (Mark-

Oliver Würtz 24.03.2021). The result of this interview, i.e. the organizational-procedural and IT integration on the basis of the ITVU technical architecture, is presented in the graphic below. This section will finish with an explanation and a conclusion.

In the context of the study, the following two questions were asked about integration:

Is a sub-service provider involved in the provision of the BerLKönig service and if so, in what form and to what extent does the provider support BVG?

To what extent is the IT of the BerLKönig integrated into the structures of the BVG?



Fig. 4. BerLKönig: Professional integration and use of sub-service providers (own presentation based on the ITVU industry model (Scholz 2016))

As can be seen from the left side of the diagram, the company ViaVan² supports BVG in the provision of the BerLKönig service to a considerable extent: this support amounts to 100% almost everywhere. With the exception of offer planning and evaluation, the partner operates this service almost entirely on behalf of BVG.. Offer planning is supported by ViaVan with information and experience derived from operations - this also applies to the evaluations necessary for operations (as well as for sales and marketing campaigns).

As the right-hand side of the diagram (see Fig. 3) shows, IT integration of the individual service modules is virtually non-existent compared to operational integration. One exception is a low level of integration in the areas of service planning and marketing, which can be attributed to the common use of standard software (e.g. Microsoft Office).

In summary, the questions posed at the beginning of the section can be answered as follows: The BerLKönig service is operated to an extremely high degree by ViaVan. The technical and organizational support provided by ViaVan to BVG is very high; the IT systems used are not integrated into the IT architecture of BVG.

¹ Michael Bartnik has been in the public transport industry for 22 years and has been helping to shape the digital transformation at BVG since 2016. He leads their Jelbi project; an app that determines the shortest routes intermodally by using BVG

transport and various partners. He has extensive knowledge about the BerLKönig service.

² ViaVan is a joint venture between Daimler AG, Mercedes-Benz Vans division and Via Transportation Inc. a US-American-Israeli company.

C. Integration of NMS into a classic supply-oriented public transport company using BerlKönig as an example

In this section, the authors present the required functions with regard to an integration of an NMS service (BerlKönig) into a supply-oriented public transport company. For this purpose, they outline the simple integration of the sales area/fare management of the BerlKönig service as an example of an NMS service into the BVG structure. The envisioned solution is based on the BVG's current orientation, which follows the organizational and business model of a pipeline company. This is a first step for the development of a more comprehensive enterprise architecture that integrates the supply-platform-oriented public transport operations on the one hand and the demand- and platform-oriented modern NMS on the other hand.

In Figure 5 the individual function blocks of an overall solution are considered.

Hereby we look at the sales area and in particular the fare management, which is triggered by the upstream ticketing processes. The BerlKönig service is only offered via the app distribution channel. There is no other way to order and use this service. Therefore, the design of the app in terms of usability and availability is of great importance.

The above-mentioned functional blocks are divided into seven systems; together they form the structure of the settlement system.

In each individual system, there are application functions that are relevant for the fulfilment of the system requirements. As an example, the area of the settlement system is highlighted below.

The following application functions are important for handling the business process:

- **Activation:** After registration (agreement system), the customer still has to be activated so that he can conclude transactions on the platform.
- **Ticketing:** The travel authorization booked is expressed in a product, the ticket.
- **Payment:** The payment process runs via a payment service provider (external) who collects the amount from the customer -> passenger.
- **Settlement:** Settlement takes place between the payment service provider and the BVG on an agreed day each month. Money and information (individual and total transaction/s) are exchanged here.

Figure 6 depicts an architectural solution developed by the authors for integrating the sales and fare management areas of the BerlKönig service.

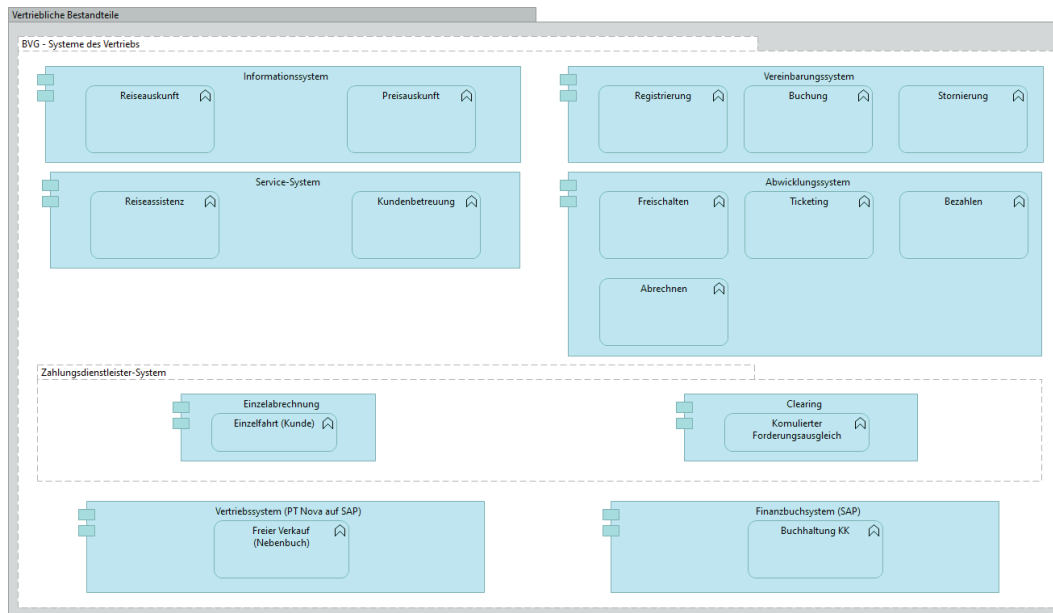


Fig. 5. Functional blocks of an NMS integration using BerlKönig as an example(own representation)

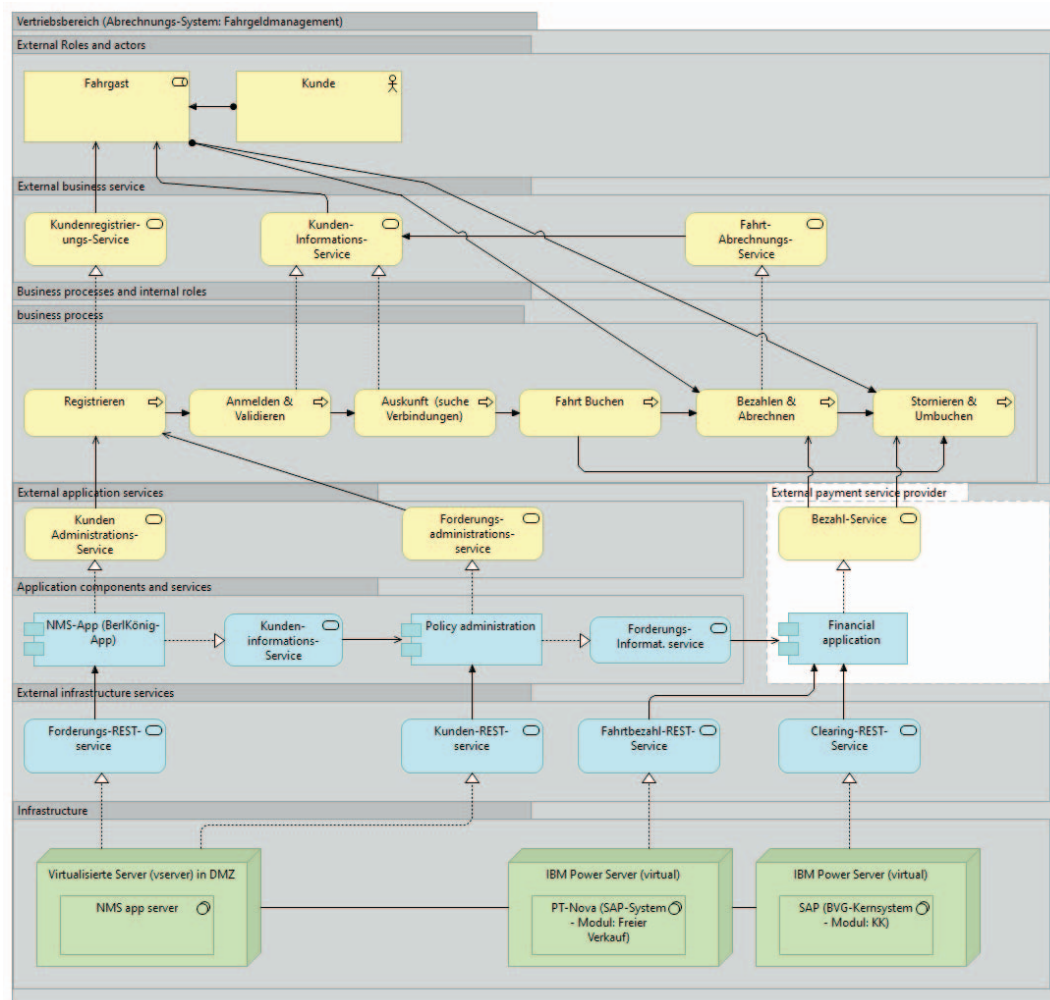


Fig. 6. Target architecture for the integration of the sales area/fare management of the BerlKönig service (own representation)

Hereby, the focus is on receivables management. The infrastructure used is - apart from the background software of the app - identical to the one used by the BVG. This means that three software systems are used on the hardware mentioned:

- PT-Nova (Hansecom) on SAP as a sub-ledger for the distribution channel,
- NMS app Server (background software) as the central application for the app,
- BVG's SAP core system for billing the payment service provider (clearing).

Based on these central systems, the authors then modelled a modern architecture that encapsulates the individual levels through standardized services. Compared to the functional requirements, the non-functional requirements played no role in the current architecture.

For an initial evaluation of the architecture approach, we use established quality attributes for enterprise architectures, as for example defined in [21]. The intention of the architecture draft is to support the integration of NMS in the areas of claims and fare management. The architecture is supposed to take into account the existing software application of the BVG, which has to be central to the evaluation. Table 1 shows the quality criteria and our impression to what extent they were covered.

TABLE I. INITIAL EVALUATION OF THE ARCHITECTURE

Quality Attribute	Evaluation
Platform independence	Assessment: partly met Coverage of the most important applications and functions by services.
Scalability	Assessment: fully met Due to the structured distributed structure, it is possible to achieve a high scalability.

Flexibility	Assessment: partly met Flexibility is provided by orchestrating appropriate services.
Reliability	Assessment: shortcunning
Component reusability	Assessment: fully met High reusability through the use of services.
Interoperability	Assessment: fully met The different applications are served by standard interfaces of the services.
Simplicity	Assessment: partly met Simplification is only possible to a limited extent in architecture.
Monitoring, auditing, logging	Assessment: fully met Can be fully implemented in architecture.
Security	Assessment: fully met Security can be specifically considered in this distributed architecture.

The individual quality criteria presented in Table 1 provide information about the integration capability of the architecture. The service orientation aimed for in the design enables the architecture to adopt a flexible structure, whereby its functions can be provided by various applications.

In conclusion, it can be said that this first draft of an appropriate architecture can be integrated well into existing infrastructures - or, in particular, into those of the BVG - due to its service orientation. However, this initial architecture version needs further development and evaluation outside the context of BVG.

VI. SUMMARY, CONCLUSION AND FURTHER PUBLICATION

In the following, the authors briefly summarize the findings as well as the questions posed in the paper and their answers with additional comments:

The (second) objective formulated in the introduction to present a (simple) integration of the sales/fare management into the "normal" architecture of a pipeline company was fulfilled in the (last) section "Integration of NMS into a classic supply-oriented public transport company using BerlKönig as an example". Thereby, the integration of sales/fare management was a solvable task under the above-mentioned conditions. A future challenge is the strategic alignment of supply and demand oriented services within the public transport company - this was not the task in this working paper and will be discussed in a later publication.

In the section "Supply- and demand-oriented or platform- and pipeline-oriented public transport services", the general differences between both approaches were compared for a better understanding and it was confirmed in the following section ("Comparison between the functional requirements of demand- and supply-oriented public transport services") that both approaches can lead to different requirements for the enterprise architecture.

The two questions from the section "Professional and IT integration of BerlKönig" were answered as follows: The BerlKönig service is operated to an extremely high degree by ViaVan. The technical and organizational support rendered by ViaVan to BVG is very high; the IT systems used are only rudimentarily integrated into the IT architecture of BVG. This finding confirms the above-mentioned objective set by the authors.

Finally, in the section "Integration of NMS into a classic supply-oriented public transport company using the example of the BerlKönig service", the authors have presented the required functional blocks and developed a target architecture for the integration of the sales area/fare management based on those functional blocks and the ITVU industry model.

The following conclusion can be drawn based on the comments above, the findings and questions answered:

Introducing the functions of NMS into an existing supply-oriented public transport company is a task that can be solved as described herein. The challenge lies not in integrating the individual functional blocks presented in the previous chapters into the company, but in dealing with the strategic orientation of both approaches.

It does not seem to make much sense to substitute supply-orientation with demand-orientation, as public transport has basically proven itself with the number of passengers it services in the context of supply-orientation. It will be a matter of supporting the supply-oriented services of public transport with demand orientation. For this purpose, it is important to promote and coordinate the interaction of both approaches.

In addition to the architectural orientation, the focus will be on the depth of documentation, i.e. the question: what must and what should be recorded in the enterprise architecture in order to carry out a qualitative assessment of both approaches.

In conclusion, it can be confirmed that the (first) objective set out in the introduction,

"to make a scientific contribution to the field of enterprise architecture management in public transport companies", was fulfilled.

VII. FUTURE WORK

In future work, the authors intend to develop the central requirements for a platform-oriented business model operating in a supply-oriented public transport company. Both functional requirement blocks (demand and supply orientation) are to be taken into account in a solution architecture. The aim is to develop a new architecture that shows how such an architecture could function in the future as a central broker in the context of both approaches.

This future work aims to identify and present in an orderly manner all relevant requirements needed for modelling an artefact in the form of an enterprise architecture.

In addition to the above requirements, there are other requirements that need to be considered in a comprehensive architecture. On the one hand, requirements of functions that are only relevant to one of the approaches (pipeline or supply vs. platform or demand) have to be considered and on the other hand, it has to be decided how to deal with conflicting requirements resulting from both approaches.

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